Chapter summary

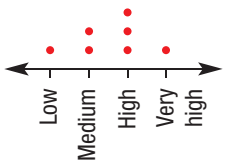
Probability and statistics

Data

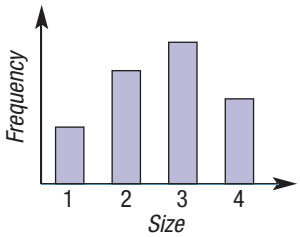
Categorical	Numerical
• Nominal (e.g. red, blue, ...)	• Discrete (e.g. 1, 2, 3, ...)
• Ordinal (e.g. low, medium, ...)	• Continuous (e.g. 0.31, 0.481, ...)

Graphs for a single set of categorical or discrete data

Dot plot



Column graphs



Stem-and-leaf plot

Stem	Leaf
0	1 6
1	2 7 9
2	3 8
3	4

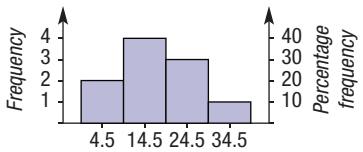
2 | 3 means 23

Frequency tables

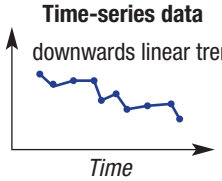
Class interval	Frequency	Percentage frequency
0–9	2	20%
10–19	4	40%
20–29	3	30%
30–39	1	10%
Total	10	100%

$$\text{Percentage frequency} = \frac{\text{frequency}}{\text{total}} \times 100$$

Histogram



Time-series data



Measures of centre

- Mean ($\bar{x}$) = $\frac{\text{sum of all values}}{\text{number of values}}$
- Median = middle value of ordered data

odd number	even number
1 4 ⑥ 9 12	1 2 4 6 7 11
↑ median	Median = $\frac{4+6}{2} = 5$

- Mode = most common value

Measure of spread

- Range = max – min

Probability

- Sample space is the list of all possible outcomes
- $P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$

Notation

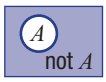
- A or B



- A and B



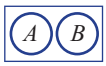
- Complement of A is not A.



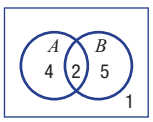
- A only



- Mutually exclusive events



Venn diagram



Two-way table

	A	not A	
B	2	5	7
not B	4	1	5
	6	6	12

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Chapter review

Chapter tests and worksheets can be found in the Teacher Resource Package. Interactive fill-in-the gaps and drag & drop literacy activities can be found in the Interactive Textbook.

Multiple-choice questions

An online version of this test is available in the Interactive Textbook.

1 A letter is chosen from the word SUCCESS. The probability that the letter is not a C is:

A $\frac{2}{7}$ **B** $\frac{3}{5}$ **C** $\frac{5}{7}$ **D** $\frac{4}{7}$ **E** $\frac{3}{7}$

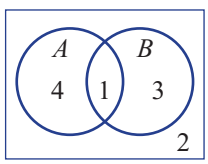
2 The number of manufacturing errors spotted in a car plant on 20 days is given by this table.

Number of errors	0	1	2	3	Total
Frequency	11	6	2	1	20

An estimate of the probability that on the next day no errors will be observed is:

A $\frac{3}{10}$ **B** $\frac{9}{20}$ **C** $\frac{11}{20}$ **D** $\frac{17}{20}$ **E** $\frac{3}{20}$

3 For this Venn diagram, $P(A \text{ or } B)$ is equal to:



A $\frac{4}{5}$ **B** $\frac{1}{2}$ **C** $\frac{5}{8}$ **D** $\frac{1}{4}$ **E** $\frac{1}{10}$

4 15 people like apples or bananas. Of those 15 people, 10 like apples and 3 like both apples and bananas. How many people from the graph like only apples?

A 5 **B** 3 **C** 13 **D** 7 **E** 10

5 For this two-way table, $P(A \text{ and } B)$ is:

A $\frac{2}{3}$ **B** $\frac{1}{4}$ **C** $\frac{1}{7}$ **D** $\frac{1}{3}$ **E** $\frac{2}{7}$

	A	not A	
B		1	3
not B			4
		4	

6 What type of data are generated by the survey question: 'What is your favourite sport to play?'

A numerical and discrete
B numerical and continuous
C categorical and continuous
D categorical and nominal
E categorical and ordinal

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Questions **7** and **8** refer to the stem-and-leaf plot below.

Stem	Leaf
2	4 9
3	1 1 7 8
4	2 4 6
5	0 4

4|2 means 42

7 The minimum score in the data is:
A 4 **B** 0 **C** 24 **D** 38 **E** 54

8 The mode is:
A 3 **B** 31 **C** 4 **D** 38 **E** 30

9 The range and mean of 2, 4, 3, 5, 10 and 6 are:
A range = 8, mean = 5
B range = 4, mean = 5
C range = 8, mean = 4
D range = 2–10, mean = 6
E range = 8, mean = 6

10 The median of 29, 12, 18, 26, 15 and 22 is:
A 18 **B** 22 **C** 20 **D** 17 **E** 26

Short-answer questions

1 A fair 6-sided die is rolled once. Find:
a $P(4)$ **b** $P(\text{even})$ **c** $P(\text{at least 3})$

2 A letter is chosen from the word INTEREST. Find the probability that the letter will be:
a I **b** E **c** a vowel
d not a vowel **e** E or T

3 An engineer inspects 20 houses in a street for cracks. The results are summarised in this table.

Number of cracks	0	1	2	3	4
Frequency	8	5	4	2	1

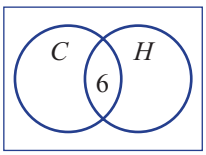
a From these results, estimate the probability that the next house inspected in the street will have the following number of cracks:
i 0 **ii** 1 **iii** 2 **iv** 3 **v** 4
b Estimate the probability that the next house will have:
i at least 1 crack
ii no more than 2 cracks

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Chapter review

4 Of 36 people surveyed, 18 have an interest in cars (C), 11 have an interest in homewares (H) and 6 have an interest in both cars and homewares.

a Complete this Venn diagram.



b Complete this two-way table.

	C	not C	
H	6		
not H			

c State the number of people surveyed who do not have an interest in either cars or homewares.

d If a person is chosen at random from the group, find the probability that the person will:

i have an interest in cars and homewares

ii have an interest in homewares only

iii not have any interest in cars

5 All 26 birds in an aviary have clipped wings and/or a tag. In total, 18 birds have tags, 14 have clipped wings and 6 have both clipped wings and a tag.

a Find the number of birds that have only clipped wings.

b Find the probability that a bird chosen at random will have a tag only.

6 A group of 16 people was surveyed to find the number of hours of television they watch in a week. The raw data are listed:

6, 5, 11, 13, 24, 8, 1, 12,
7, 6, 14, 10, 9, 16, 8, 3

a Organise the data into a table with class intervals of 5. Start with 0–4, 5–9 etc. Include a tally, frequency and percentage frequency column.

b Construct a histogram for the data, showing both the frequency and percentage frequency on the graph.

c Would you describe the data as symmetrical or skewed?



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7 A basketball team scores the following points per match for a season.
20, 19, 24, 37, 42, 34, 38, 49, 28, 15, 38, 32, 50, 29

a Construct an ordered stem-and-leaf plot for the data.

b Describe the distribution of scores.



8 For the following sets of data, determine:

i the mean **ii** the range **iii** the median

a 2, 7, 4, 8, 3, 6, 5

b 10, 55, 67, 24, 11, 16

c 1.7, 1.2, 1.4, 1.6, 2.4, 1.3

9 Thirteen adults compare their ages at a party. They are:
40, 41, 37, 32, 48, 43, 32, 76, 29, 33, 26, 38, 87

a Find the mean age of the adults, to 1 decimal place.

b Find the median age of the adults.

c Why do you think the mean age is larger than the median age?

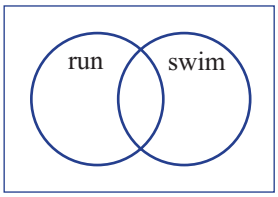


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Chapter review

Extended-response questions

1 Of 15 people surveyed to find out if they run or swim for exercise, 6 said they run, 4 said they swim and 3 said they both run and swim.





a How many people surveyed neither run nor swim?

b One of the 15 people is selected at random.
Find the probability that they:
i run or swim **ii** only swim

c Represent the information in a two-way table.

2 The number of flying foxes taking refuge in a fig tree is recorded over a period of 14 days. The data collected are given here.

73, 50, 36, 82, 15, 24, 73, 57, 65, 86, 51, 32, 21, 39

a Arrange the data in ascending order.

b Find the:
i mean
ii median
iii range

c Describe the distribution. Give two possible reasons why the numbers of flying foxes taking refuge varies so much.



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